Charles Brandon Duplantier

Bioinformatics & Research Data Analyst | Python · SQL · reproducible, externally-validated analysis

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SUMMARY

Data analyst completing a B.S. in Bioinformatics (University of Arizona, expected December 2027) who builds reproducible statistical analyses on biomedical and operational data. Reports confidence intervals, sample-size limits, and external-validation figures rather than the most favorable number. Independent work spans epigenetic aging clocks with external validation, survival and clinical-outcome modeling, and RNA-seq analysis; current internship builds Python reporting systems for a search-and-rescue organization.

TECHNICAL SKILLS

Languages & data: Python (pandas, NumPy, SciPy, scikit-learn), R, SQL, PostgreSQL, DuckDB, Snowflake
Bioinformatics / stats: pydeseq2 (RNA-seq differential expression), lifelines (survival analysis), logistic regression, external cross-cohort validation, GEO datasets
BI & tooling: Power BI (DAX), Tableau, Looker Studio, Excel, Git, Flask, Streamlit, REST APIs, Claude API, Google Analytics

EXPERIENCE

Data Analyst & Integration Developer (Internship), Nalaquq, LLC / Quinhagak Search and Rescue — Remote · May 2026 to Present

- Built a search-coverage and probability-of-detection model that converts searcher GPS tracks into a map of ground actually covered versus never walked, using published SAR literature, with conservative sweep-width defaults so coverage is understated rather than overstated; demonstrated on simulated GPS tracks.
- Built an operational reporting pipeline for callout history (seasonal timing, village activity, outcome, search duration) with a sample-size gate that refuses to report patterns below n=10 and a privacy guard that coarsens any count under 5 so small numbers cannot identify individuals; tested on simulated records, with real-data mode blocked until the schema is confirmed.
- Built a CASEVAC (casualty evacuation) reporting bridge that restored casualty-report submission for iPhone users who could not access the existing feature.
- Wrote nine test modules across five feature branches; a temporary branch combining all five passed all 240 tests.

Independent Data Analysis, Duplantier Advertising (self-directed practice) — Dayton, OH · 2018 to Present

- Built a Python lead-sourcing pipeline on the Google Places API covering 1,037 cities across 38 states, running daily with deduplication, to source and score small-business prospects (HVAC, plumbing, electrical, dental, and auto-repair target markets).
- Built a Google Sheets CRM in Apps Script (lead tracking and pipeline reporting) and a Flask site-audit tool that scores prospect websites 0 to 100 using the Claude API.
- Ran web analytics and reporting for a local alterations business (the practice's one external engagement): built and maintain the site and used Google Analytics to time content against seasonal demand; annual site visits grew from 927 (2018) to 3,020 (2025).

Pathologist Assistant, Montgomery County Coroner's Office — Dayton, OH · 2010 to 2013

- Produced precise anatomical documentation in a high-volume clinical setting under strict protocol and quality standards.

Technical Support Specialist, Geek Squad — 2013 to 2014

- Device diagnostics, OS troubleshooting, and customer data recovery.

SELECTED ANALYTICS PROJECTS

Independent, self-directed analyses on public datasets.

- **DNA methylation aging clock:** externally validated on an independent cohort (GSE42861, 689 samples), MAE 6.36 years (r = 0.884); training-set test MAE 5.65 (r = 0.892). Trained on GSE40279 (656 whole-blood samples), 176 CpG sites selected from about 20,000.
- **Hospital 30-day readmission model:** logistic regression on a 300-patient synthetic CMS-HRRP-modeled dataset. Heart failure carried 19.7 times the odds of readmission versus hip/knee replacement (95% CI 5.3 to 72.9, p < 0.001); scheduled follow-up was associated with lower odds (OR 0.58, p = 0.09, not statistically significant). Random forest ROC AUC 0.587 on a held-out test set.
- **Medicare HRRP measurement analysis:** across 18,330 records from 3,055 hospitals, found 36.1% of records unscored because a hospital needs 25 qualifying patients per condition, a gap that falls hardest on small hospitals.
- **Biological-age mortality analysis:** 28,510 U.S. adults followed up to 19 years; each additional year of biological age was associated with about 4.4% higher mortality hazard (HR 1.044).
- **Brain RNA-seq:** differential-expression analysis identifying 119 age-linked genes.

EDUCATION & CERTIFICATIONS

- **B.S. Bioinformatics**, University of Arizona, expected December 2027
- Google Data Analytics Professional Certificate (2024); The Global Career Accelerator (2025): Python and Data, Querying Data, AI Professional Skills, Intercultural Skills; Skydio X10 Professional Operator Training (2026, manufacturer course, not FAA Part 107).